BACT7507 – Week 3 Lab

Solution Design & Proposal Draft Development

Project Title: Smart Cycling Route

Part 1 – Functional Requirements

Core Features

1. Cycling Route Search and Planning

Description:

This system will allow users to enter their starting location and destination. The app will generate a cycling route between the two locations and display the distance and estimated cycling time.

Example:

Start:
Auckland City Centre

Destination:
Ponsonby

The system will show the recommended cycling route all the way

2. Cycle Lane and Road Type Visualisation

Description:

The system will display different road types using different colours on the map. This helps cyclists identify which areas are suitable for cycling.

The map will show:

- Green: Dedicated cycle lanes
- Blue: Shared cycle paths
- Yellow: Bus lanes
- Grey: Normal vehicle roads

This feature helps users understand the type of road they will use before starting their journey.

3. Route Safety Rating and Recommendation

Description:

The system will analyse the selected route and provide a cycling suitability rating.

Example:

Route Safety Rating:

Highly Recommended

Cycle Lane Coverage:
80%

Normal Road:
20%
The system will recommend routes with more cycling infrastructure.

4. Turn-by-Turn Route Guidance

Description:

The system will provide basic navigation instructions to help users follow the selected route.

Example:

1. Start from Auckland CBD

2. Turn left onto Nelson Street

3. Continue on protected cycle lane

4. Turn right towards Ponsonby Road

5. User Account and Saved Routes

Description:

Users will be able to create an account and save frequently used cycling routes.

Examples:

- Home to University
- Work to Home

This allows users to access previous routes easily.

Non-Functional Requirements

1. Usability

The application should have a simple and clear interface. Users should easily understand the map colours, route information, and cycling recommendations without technical knowledge.

2. Performance

The system should load maps and display route information within a reasonable time after the user enters locations.

3. Security

User account details should be stored securely. Password information should be protected from unauthorised access.

Part 2 – Scope Confirmation

Features Included

The project will include:

A. Search starting location and destination

B. Display cycling routes on an Auckland map

C. Highlight cycle lanes, shared paths, bus lanes, and normal roads

D. Display distance and estimated cycling time

E. Provide basic turn-by-turn directions

F. Provide route safety ratings

G. Allow users to save favourite routes

Features NOT Included

The following features are outside the project scope:

- Real-time GPS tracking

- Voice navigation
- Live traffic updates
- Automatic route changes while travelling
- Weather forecasting
- Social media features
- Bike rental booking
- Payment system
- Full replacement of Google Maps

Part 3 – System Design Diagram

Block Diagram

(Create this using Draw.io and upload the image/PDF to GitHub)

USER

User Interface |
| HTML/CSS/JavaScript|
|

| Application Logic |
Python Backend

| MySQL | | Map Service |
| Database | | OpenStreetMap|

User Flowchart

START

Open CycleSafe Auckland

Enter Start Location

Enter Destination

Select Route Preference

(Safest / Fastest)

System Processes Request

Retrieve Map and Cycle Lane Data

Display Route

Show:

- Distance
Time
Cycle Lane Information
Safety Rating

Provide Directions

Save Route (Optional)

END

Part 4 – Technology Selection & Justification

Programming Language and Framework

Python

Justification:

Python will be used as the backend framework because it is lightweight, simple to learn, and suitable for developing a small web application. Python can handle user requests, process route information, and communicate with the database.

Frontend Technologies

HTML, CSS, JavaScript

Justification:

HTML, CSS, and JavaScript will be used to develop the user interface. HTML creates the webpage structure, CSS provides design and layout, and JavaScript enables interactive features such as map controls and user input handling.

Database

MySQL

Justification:

MySQL will be used to store user information, saved routes, and application data. It is reliable, widely used, and suitable for managing structured information.